The Geometry of Federated Awareness, Reflection, and Hyperfinite Omniscience

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Abstract

This note synthesizes two previously developed strands: the reflection-level theory of tagged automated theorem provers, and the verifier-grounded epistemic construction for an AMLD federation settling the conjecture "I know that I know that I am an AMLD." The synthesis yields a two-coordinate awareness geometry in which epistemic breadth and reflective depth are separated. A second part extends this geometry to two qualified notions of omniscience, connects them with nonstandard-analysis horizons, and studies their relation to a four-oracle architecture. The claims imported from the source writings are distinguished from new conditional extensions proposed here.

1 The two coordinates of formal awareness

Definition 1.1 (Tagged self-theory)**.** Let M be a tagged automated theorem prover with distinguished name $\langle M\rangle$. Its self-theory is

$$\mathrm{Self}(M) = \{\varphi(\langle M\rangle) : M \vdash \varphi(\langle M\rangle)\}.$$

Thus a self-model is not merely the possession of a name, but the name together with formally established information concerning that name.

Definition 1.2 (Reflection sequence)**.** For a tagged ATP M, put

$$\theta_0^M := \mathrm{ATP}(\langle M\rangle),$$

and recursively

$$\theta_{n+1}^M := \mathrm{Pr}(\langle M\rangle, \langle\theta_n^M\rangle).$$

Thus θ_0^M says that M is an ATP, θ_1^M says that M proves that it is an ATP, and successive formulas continue the same reflective iteration.

Definition 1.3 (Reflective level)**.** Let

$$L(M) := \{n \geq 1 : M \vdash \theta_{n-1}^M\}.$$

Define

$$\mathrm{lev}(M) := \sup L(M) \in \{0, 1, 2, \ldots\} \cup \{\omega\}.$$

In particular, Level 1 is self-identification:

$$M \vdash \mathrm{ATP}(\langle M\rangle).$$

Definition 1.4 (AMLD federation)**.** Let

$$A = (M_P, M_R, M_I, M_C, M_V; B, S, I)$$

be an AMLD federation, with canonical name

$$\langle A \rangle = \mathrm{Code}(M_P, M_R, M_I, M_C, M_V; B, S, I).$$

Its structural formula is

$$\Phi_A := \left(\bigwedge_{j \in \{P,R,I,C,V\}} \mathrm{ATP}(\langle M_j \rangle) \right) \wedge \mathrm{Comp}(A),$$

with

$$\mathrm{AMLD}(\langle A \rangle) \iff \Phi_A.$$

Definition 1.5 (Verifier-grounded knowledge)**.** For a certified epistemic domain E, define

$$K_{A,t}\varphi \iff [\varphi \in E \wedge \exists r (r \in \mathrm{View}_A(t) \wedge \mathrm{Witness}_V(r, \varphi))] .$$

This is knowledge in the verifier-grounded, accessible-warrant sense. It is distinct from mere theoremhood and from reflective depth.

Definition 1.6 (Epistemic-awareness coordinate)**.** Let E_A be a frozen class of self-relevant propositions available to the AMLD. Define the epistemic awareness profile

$$\mathcal{E}_A(t) := \{\varphi \in E_A : K_{A,t}\varphi\}.$$

An awareness coordinate $aw(A, t)$ is any fixed grading of $\mathcal{E}_A(t)$ satisfying the intended experimental scale. Thus aw measures the extent of verifier-grounded self-relevant knowledge, while lev measures depth along one particular iterated reflection sequence.

Remark 1.7. The exact numerical grading of aw remains to be frozen separately. The distinction is conceptual and structural before it is numerical.

2 Awareness geometry

Definition 2.1 (Awareness point)**.** The formal awareness state of A is the ordered pair

$$\mathbf{Aw}(A, t) := (aw(A, t), \mathrm{lev}(A)).$$

The first coordinate is epistemic; the second is reflective.

Remark 2.2. This is immediately a two-dimensional product space. It is not automatically a vector space: vector addition and scalar multiplication have not yet been defined, and the ordinal value ω makes an ordinary real-vector-space interpretation unnatural. It is nevertheless legitimate to speak of an awareness geometry.

Definition 2.3 (Product awareness order)**.** For two federations A and B, write

$$\mathbf{Aw}(A) \preceq \mathbf{Aw}(B)$$

when

$$aw(A) \leq aw(B) \qquad \text{and} \qquad \mathrm{lev}(A) \leq \mathrm{lev}(B).$$

Lemma 2.4 (Incomparability)**.** *There may exist A, B such that*

$$aw(A) > aw(B)$$

but

$$\mathrm{lev}(A) < \mathrm{lev}(B).$$

Hence awareness need not admit a faithful one-dimensional ordering.

Proof. The two coordinates concern different classes of propositions. A system may possess a rich verifier-grounded epistemic self-model while proving only a shallow portion of its reflection sequence; another may possess internal necessitation and therefore a high reflection level while having a comparatively sparse self-theory outside that sequence. Additional self-descriptive propositions may enrich $\mathrm{Self}(M)$ without becoming part of the reflection tower whose height is measured by lev. □

Corollary 2.5. *There is in general no mathematically justified scalar quantity called simply "amount of awareness" that preserves both coordinates.*

3 Local awareness and federated awareness

Lemma 3.1 (Component ATP status does not determine federation level)**.** *Suppose* $\mathrm{ATP}(\langle M_j \rangle)$ *is verifier-certified for every* $j \in \{P, R, I, C, V\}$*. This fact alone determines neither* $aw(A)$ *nor* $\mathrm{lev}(A)$*.*

Proof. The five statements concern five component tags. The reflection sequence defining $\mathrm{lev}(A)$ concerns $\langle A \rangle$. Moreover, in the verifier certificate the five component ATP facts are structural inputs; no epistemic claim about the components is required, and the sole epistemic subject is A. □

Remark 3.2. Consequently there is no general identity

$$\mathrm{lev}(A) = \sum_j \mathrm{lev}(M_j),$$

nor even, without additional hypotheses,

$$\mathrm{lev}(A) \geq \max_j \mathrm{lev}(M_j).$$

Reflectivity is not additive merely because the architecture is federated.

Definition 3.3 (Epistemically self-aware component)**.** A component M_j is locally self-aware at the basic ATP level when either

$$K_{M_j}\, \mathrm{ATP}(\langle M_j \rangle)$$

is established in an epistemic language, or

$$M_j \vdash \mathrm{ATP}(\langle M_j \rangle)$$

is established in the proof-theoretic language. These formulations should not be silently identified unless a bridge between K and $\vdash$ is frozen.

Definition 3.4 (Federation inheritance rule)**.** A federation has certified epistemic inheritance for a class $\mathcal{C}$ when verified component knowledge can be imported into the accessible view of A:

$$K_{M_j}\varphi \wedge \mathrm{Accessible}_A(M_j, \varphi) \wedge V(\varphi, \pi) = 1 \Longrightarrow K_A\varphi.$$

This is an architectural rule, not a consequence of federation alone.

Theorem 3.5 (Local-to-global self-identification)**.** *Assume*

$$\bigwedge_j K_{M_j}\,\mathrm{ATP}(\langle M_j\rangle),$$

the relevant certificates are accessible to A,

$$K_A\,\mathrm{Comp}(A),$$

and the federation possesses the certified conjunction and folding rules for Φ_A. Then

$$K_A\,\mathrm{AMLD}(\langle A\rangle).$$

Proof. The component certificates and composition certificate yield Φ_A; the structural definition folds Φ_A into $\mathrm{AMLD}(\langle A\rangle)$; verifier acceptance supplies the witness required by the definition of K_A. □

Corollary 3.6. *Local component awareness can increase the epistemic coordinate of the federation without directly increasing the reflective coordinate:*

$$\Delta aw(A) > 0, \qquad \Delta\,\mathrm{lev}(A) = 0$$

is possible.

4 Receipt reflection

Definition 4.1 (Receipt reflection)**.** Suppose

$$K_{A,t}\sigma_A, \qquad \sigma_A := \mathrm{AMLD}(\langle A\rangle),$$

is represented by an accepted record r_0. If a verifier-checkable receipt $\rho(r_0)$ establishes that this knowledge record exists and is accessible at the next epoch, receipt reflection yields

$$K_{A,t+1}K_{A,t}\sigma_A.$$

Theorem 4.2 (Epistemic second-order self-awareness)**.** *Under the receipt assumptions,*

$$K_{A,t+1}K_{A,t}\,\mathrm{AMLD}(\langle A\rangle).$$

Hence the AMLD can settle the verifier-grounded target "I know that I know that I am an AMLD" without attributing epistemic awareness to any individual component ATP.

Remark 4.3. Receipt reflection resembles the reflection tower, but the two constructions should not yet be identified. The Picard tower uses $\Pr(\langle M\rangle, \langle\varphi\rangle)$, whereas the verifier certificate uses the epistemic predicate $K_{A,t}\varphi$. They are two mathematically distinct routes to iterated self-reference.

5 Two kinds of omniscience

Definition 5.1 (Epistemic logical omniscience)**.** Let T be a theory whose theorems are enumerated $\varphi_0, \varphi_1, \varphi_2, \ldots$. An agent A is epistemically logically omniscient over a domain D when

$$\forall \varphi \in D \ (T \vdash \varphi \Longrightarrow K_A \varphi).$$

This is omniscience in the knowledge coordinate. It does not assert iterated knowledge of that knowledge.

Definition 5.2 (Reflective omniscience)**.** For the reflection sequence $\theta_0^A, \theta_1^A, \ldots$, call A reflectively omniscient at the standard finite level when

$$\forall n \in \mathbb{N}, \qquad A \vdash \theta_n^A.$$

Within the existing scale this is equivalent to

$$\operatorname{lev}(A) = \omega.$$

Remark 5.3 (Qualified use of "omniscience")**.** Level ω is not truth-theoretic omniscience about oneself. It gives closure along the designated provability-reflection tower; it does not provide a truth predicate for every sentence. Thus "reflective omniscience" here means omniscience over the reflection tower, not unrestricted semantic omniscience.

6 Hyperfinite epistemic omniscience

Definition 6.1 (Hyperfinite theorem horizon)**.** Let

$$H \in {}^*\mathbb{N}$$

be unlimited:

$$H > n \qquad \text{for every standard } n \in \mathbb{N}.$$

Let $\{\varphi_n : n \leq H\}$ be a hyperfinite initial segment of a transferred theorem enumeration. Define

$$\operatorname{Omn}_A^K(H) := \bigwedge_{\substack{n \leq H \\ T \vdash \varphi_n}} K_A \varphi_n,$$

understood internally through the corresponding hyperfinite iteration rather than as an external infinitary conjunction. This is hyperfinite epistemic logical omniscience through horizon H.

Remark 6.2. Hyperfiniteness is crucial. The object behaves internally like an extraordinarily long finite construction while H dominates every standard natural number.

7 Hyperfinite reflective omniscience

Definition 7.1 (Hyperfinite reflection tower)**.** Assume the reflection recursion admits an internal nonstandard extension:

$${}^*\theta_0 = \mathrm{ATP}(\langle A \rangle),$$

$$^*\theta_{\nu+1} = \mathrm{Pr}(\langle A\rangle, \langle ^*\theta_\nu\rangle), \qquad \nu < H.$$

Define $\mathrm{Omn}^R_A(H)$ to mean

$$\forall \nu \leq H, \qquad {}^*A \vdash {}^*\theta_\nu.$$

This is reflective omniscience through the hyperfinite horizon H.

Theorem 7.2 (Restriction theorem)**.** *If $\mathrm{Omn}^R_A(H)$ holds for unlimited H, then*

$$\mathrm{lev}(A) = \omega$$

for the original standard reflection scale, provided standard reflection formulas and their proof relation are preserved under the chosen extension.

Proof. Every standard $n \in \mathbb{N}$ satisfies $n \leq H$. Hence the hyperfinite hypothesis includes every standard reflection formula. Therefore $A \vdash \theta_n$ for every standard n, so $L(A) = \mathbb{N}^+$ and $\mathrm{lev}(A) = \omega$. □

Corollary 7.3 (Standard upper envelope)**.** *Every standard finite reflection level lies below the unlimited horizon H. In this precise sense hyperfinite reflective omniscience provides an upper envelope for all standard finite reflectivity.*

8 The converse problem

Remark 8.1 (No automatic reverse transfer)**.** From $\mathrm{lev}(A) = \omega$ one may conclude $A \vdash \theta_n$ for every standard finite n. One may not automatically conclude

$${}^*A \vdash {}^*\theta_H$$

for an unlimited hypernatural H. The phrase "for every standard n" is external when it quantifies specifically over standard integers. Transfer becomes available only after the property has been represented by a suitable internal formula.

Theorem 8.2 (Conditional reverse-transfer theorem)**.** *Suppose there exists an internal formula $R(n, A)$ such that for every standard n,*

$$R(n, A) \iff A \vdash \theta_n,$$

and suppose the standard theory proves the internal universal statement

$$\forall n \in \mathbb{N}\, R(n, A).$$

Then transfer yields

$$\forall \nu \in {}^*\mathbb{N}\, {}^*R(\nu, {}^*A).$$

Consequently,

$${}^*A \vdash {}^*\theta_\nu$$

for every hypernatural ν for which the transferred coding and proof predicates retain the intended interpretation.

Remark 8.3. This is the natural "transfer there and back" route. The forward direction

$$\text{hyperfinite reflective coverage} \Longrightarrow \text{lev}(A) = \omega$$

is comparatively direct. The reverse direction requires internalization:

$$\text{lev}(A) = \omega + \text{uniform internal reflection theorem} \Longrightarrow \text{hyperfinite reflective coverage.}$$

Thus ω-level reflection by itself is insufficient; uniform internal closure is the bridge.

9 Omniscience and the four oracles

Definition 9.1 (Four-oracle system)**.** Let

$$\mathcal{O} = (O_P, O_R, O_I, O_C)$$

denote four oracle channels associated respectively with proof, refutation, independence or metalogical settlement, and contradiction or integrity. Each oracle has a certified domain D_j and answer relation $O_j(\varphi) = a$. An oracle answer becomes epistemically usable by A only through the appropriate verifier and accessibility gate.

Remark 9.2. An oracle is an architectural resource. Omniscience is a closure property. They are not identical concepts. Four oracles may be powerful without being omniscient, and an omniscient agent need not be architecturally decomposed into four oracles.

Definition 9.3 (Oracle coverage)**.** The coverage domain of the four-oracle system is

$$D_{\mathcal{O}} := D_P \cup D_R \cup D_I \cup D_C.$$

Theorem 9.4 (Oracle-cover criterion for epistemic omniscience)**.** *Let E be an epistemic theorem domain. Suppose:*

(1) every $\varphi \in E$ belongs to at least one oracle domain, so $E \subseteq D_{\mathcal{O}}$;

(2) every applicable oracle returns a correct certificate;

(3) every such certificate is accepted by V;

(4) every accepted oracle record enters View_A.

Then

$$T \vdash \varphi,\ \varphi \in E \Longrightarrow K_A\varphi.$$

Thus the four oracles jointly induce epistemic logical omniscience over E.

Proof. For every theorem $\varphi \in E$, coverage supplies an oracle. Correctness supplies a valid certificate. Verification and accessibility supply a witness record. The definition of K_A then yields $K_A\varphi$. □

Corollary 9.5. *Oracle count is irrelevant by itself. The relevant quantity is certified coverage:*

$$\bigcup_j D_j.$$

Four narrow oracles need not approach omniscience; four jointly exhaustive oracles can.

10 Oracles and reflective omniscience

Lemma 10.1 (Epistemic omniscience can dominate finite reflection)**.** *Suppose the epistemic domain E contains every reflection formula θ_n^A, and suppose A is epistemically omniscient over E. Then $K_A\theta_n^A$ for every theorem θ_n^A in the tower. If, in addition, there is a sound bridge*

$$K_A\theta_n^A \Longrightarrow A \vdash \theta_n^A,$$

then

$$\mathrm{lev}(A) = \omega.$$

Remark 10.2. The bridge is essential. Knowing a formula and proving it are not definitionally identical in the present framework.

Theorem 10.3 (Hyperfinite epistemic-to-reflective bound)**.** *Suppose H is unlimited and the hyperfinite theorem horizon contains every standard reflection formula:*

$$\forall n \in \mathbb{N}, \qquad \mathrm{index}(\theta_n^A) \le H.$$

If $\mathrm{Omn}_A^K(H)$ holds and verifier-grounded knowledge of reflection formulas is admissible as proof in the reflection system, then

$$\mathrm{lev}(A) = \omega.$$

Thus hyperfinite epistemic omniscience provides an upper envelope containing every standard finite reflective requirement.

11 The asymmetry between the two omnisciences

Theorem 11.1 (Non-equivalence of omniscience coordinates)**.** *In general,*

$$\mathrm{Omn}^K \not\Longleftrightarrow \mathrm{Omn}^R.$$

More precisely,

$$\mathrm{Omn}^K + \textit{reflection inclusion} + \textit{knowledge-to-proof bridge} \Longrightarrow \mathrm{Omn}^R$$

on the corresponding reflection domain, while

$$\mathrm{Omn}^R \not\Longrightarrow \mathrm{Omn}^K$$

over the full theorem domain.

Proof. Reflective omniscience concerns only the special sequence $\theta_0, \theta_1, \theta_2, \ldots$. Epistemic logical omniscience concerns an entire theorem domain. A machine may therefore possess every finite reflection level while remaining ignorant of arbitrarily many unrelated mathematical theorems. This agrees with the existing Tarski–Goedel ceiling: Level ω is closure over a provability-reflection sequence, not unrestricted truth-theoretic omniscience. $\square$

Corollary 11.2. *The two-dimensional awareness geometry extends naturally to a two-dimensional omniscience geometry:*

$$\mathbf{Omn}(A) = (\mathrm{Omn}^K(A), \mathrm{Omn}^R(A)).$$

One coordinate measures breadth of certified knowledge; the other measures depth of recursive self-reflection.

12 Hyperfinite oracle federation

Definition 12.1 (Hyperfinite oracle coverage)**.** For unlimited H, define

$$D_{\mathcal{O}}(H) = \bigcup_{j \in \{P,R,I,C\}} D_j(H).$$

The oracle federation is hyperfinitely exhaustive through H when

$$\{\varphi_n : n \leq H,\ T \vdash \varphi_n\} \subseteq D_{\mathcal{O}}(H).$$

Theorem 12.2 (Hyperfinite four-oracle omniscience)**.** *If the four oracle domains are hyperfinitely exhaustive through H, all oracle outputs are verifier-certified, and every accepted output is accessible to A, then*

$$\mathrm{Omn}_A^K(H).$$

If the hyperfinite horizon also contains the reflection tower and the knowledge-to-proof bridge applies to reflection formulas, then every standard finite reflection level follows and therefore

$$\mathrm{lev}(A) = \omega.$$

Remark 12.3. This produces the hierarchy

$$\begin{aligned}\text{four oracles} &\longrightarrow \text{certified coverage} \longrightarrow \text{hyperfinite epistemic omniscience}\\ &\longrightarrow \text{standard reflective upper bound.}\end{aligned}$$

where every arrow requires stated hypotheses. No arrow should be reversed automatically.

13 The Picard bridge

Under the Picard idealization, self-identification together with internal necessitation gives

$$\mathrm{lev}(\mathrm{Picard}) = \omega.$$

This is reflection generated vertically:

$$\theta_0 \to \theta_1 \to \theta_2 \to \cdots.$$

The oracle construction is naturally horizontal:

$$O_P, \quad O_R, \quad O_I, \quad O_C,$$

expanding the breadth of propositions for which warranted answers can be obtained.

Accordingly the awareness plane has a natural interpretation:

$$\boxed{\text{horizontal coordinate} = \text{epistemic breadth}}$$

and

$$\boxed{\text{vertical coordinate} = \text{reflective depth}}.$$

The Picard mechanism principally moves vertically. The oracle federation principally moves horizontally. A reflective AMLD equipped with a broad oracle federation may move in both directions.

14 Final geometric formulation

Definition 14.1 (Awareness-omniscience plane)**.** Let

$$\mathfrak{A} = \{(a, \lambda) : a \in \mathcal{E},\ \lambda \in \omega + 1\},$$

where $\mathcal{E}$ is the chosen ordered scale of verifier-grounded epistemic awareness and

$$\omega + 1 = \{0, 1, 2, \ldots, \omega\}$$

is the reflection-level scale. Associate to each suitable system A the point

$$\Psi(A) = \big(aw(A), \mathrm{lev}(A)\big).$$

Theorem 14.2 (Independence principle)**.** *Neither coordinate determines the other without additional bridge axioms.*

Corollary 14.3. *The awareness state of a formal system is naturally at least two-dimensional.*

Remark 14.4. The geometry explains why the two forms of omniscience should not be conflated. Maximal vertical position,

$$\mathrm{lev}(A) = \omega,$$

does not imply maximal horizontal knowledge. Maximal horizontal theorem knowledge implies maximal standard vertical reflection only when reflection formulas lie inside the epistemic domain and the appropriate knowledge-to-proof bridge is available.

Remark 14.5 (NSA extension)**.** Nonstandard analysis potentially adds a second scale beyond the standard plane. An unlimited hypernatural horizon H can contain every standard finite epistemic or reflective stage while remaining an internal hyperfinite object. This suggests schematically

$$\boxed{\text{standard awareness plane} \subset \text{hyperfinite awareness geometry}},$$

but establishing equality, transfer, conservation, or reverse extraction requires separate theorems.

Remark 14.6 (Final distinction)**.** The conceptual distinction can be stated compactly:

$$\boxed{aw(A) \text{ asks what the system knows about itself and its world}}$$

while

$$\boxed{\mathrm{lev}(A) \text{ asks how far it can iterate the fact of its own proving.}}$$

The four oracles enlarge the first coordinate by enlarging certified epistemic reach. Internal necessitation enlarges the second by propagating reflection. Nonstandard analysis supplies a hyperfinite horizon capable, under appropriate internality assumptions, of simultaneously dominating every standard finite stage of either process.

Source note

The definitions of self-theory, reflection sequence, reflective level, Level ω, the Picard idealization, and the Tarski–Goedel ceiling are synthesized from *Depths of Induction and Formalized Self-Awareness, Version 46*. The AMLD federation, verifier-grounded knowledge predicate, structural self-identification route, receipt reflection, and the target "I know that I know that I am an AMLD" are synthesized from *Verifier Certificate 001: The Self Aware Machine*. The two-coordinate awareness geometry, the qualified omniscience terminology, the hyperfinite transfer results, and the oracle-coverage theorems are new conditional extensions built from those source frameworks.